UNSW Sydney
School of Electrical Engineering & Telecommunications

FINAL EXAMINATION

Term 1, 2022

ELEC1111 – Electrical Circuit Fundamentals

TIME ALLOWED:	**160 minutes**
TOTAL MARKS:	**100**
TOTAL NUMBER OF QUESTIONS:	**5**

THIS EXAM CONTRIBUTES 45% TO THE TOTAL COURSE ASSESSMENT.

Reading time: 10 minutes.

This paper contains 6 pages.

Candidates must ATTEMPT ALL questions.

Answer each question on a **separate answer sheet**.

Marks for each question are indicated beside the question.

This paper MAY be retained by the candidate.

Print your name, student ID and question number on each answer sheet.

This is an open-book examination. The use of calculators and computers, including mathematical programs like Matlab, is permitted.

Assumptions made in answering questions should be stated explicitly.

Methods used in answering questions should be described in detail.

Candidates must work **alone** without any assistance from, or interaction with, other persons. Any form of academic misconduct may result in **failure** of the course.

Upload time: 30 min. All answers must be **uploaded** on Moodle in .pdf, .gif, .jpeg, .png, or .svg formats no later than 160 min from the time the exam opens.

QUESTION 1 [20 marks]

a. **(10 marks)** A digital multimeter (DMM) is connected to a resistive circuit as shown in Figure 1. If the input resistance of the DMM is 1 MΩ, what value will be displayed if the DMM is measuring voltage (i.e., what is the voltage across the DMM resistance)?

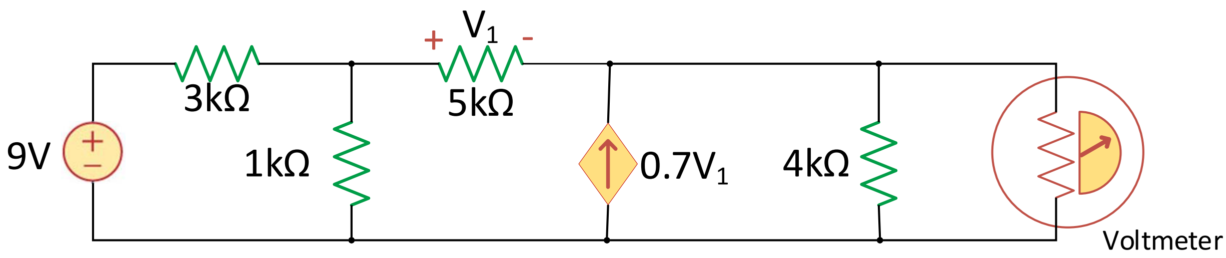

Figure 1

b. **(10 marks)** The circuit in Figure 2 represents a solar-powered waste compactor. During the day, the solar panel charges the battery. When the waste level is lower than a certain threshold (*non-compacting mode*), the switch is in position *a* and a bulb is light up. As soon as the waste level reaches the threshold (*compacting mode*), the switch moves to position *b*, a red LED turns on, and the motor starts compacting waste.

i. **(7 marks)** What is the voltage across the bulb when the waste compactor is in *non-compacting mode*?

ii. **(3 marks)** Calculate the power dissipated by the battery resistor (i.e., 0.2Ω resistor) in *non-compacting mode*.

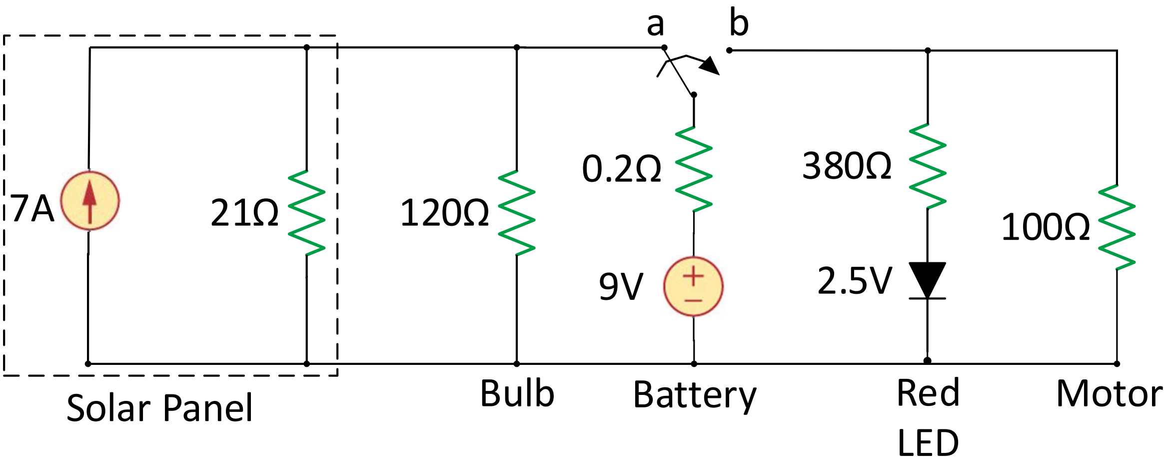

Figure 2

QUESTION 2 [20 marks]

a. **(10 marks)** The switch in the RL circuit shown in Figure 3 was closed for a long time before opening at $t = 0$ s.

i. **(8 marks)** Derive the expression for $i_0(t)$ for $t \geq 0$ s.

ii. **(2 marks)** Calculate the value of i_0 at $t = 40$ ms.

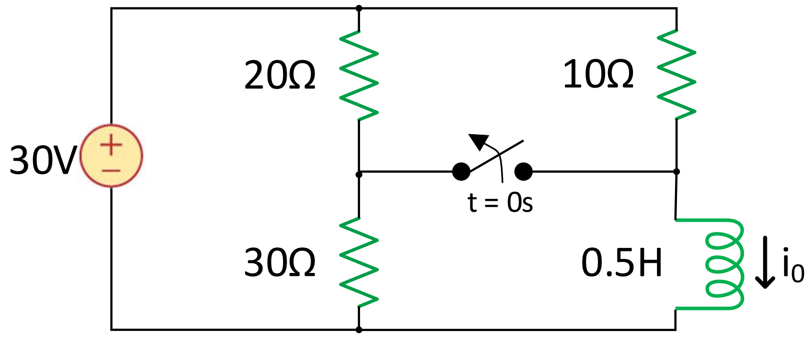

Figure 3

b. **(10 marks)** The circuit in Figure 4 shows the wiring of the headlights of a car, where the headlamp bulb has an internal resistance of 2 Ω with an inductance of 75 mH in series. Due to poor wiring, there is a 5 Ω resistance connected to the headlights circuit.

i. **(4 marks)** Find the power dissipated by the headlights (both resistor and inductor) in DC steady state conditions when the switch is closed.

ii. **(6 marks)** Taking $t = 0$ s as the time when the headlights are first connected (i.e., switch closes), derive the expression of the current across the inductor for $t \geq 0\ s$. Consider that the headlights have not been used for a long time and the inductor is initially discharged.

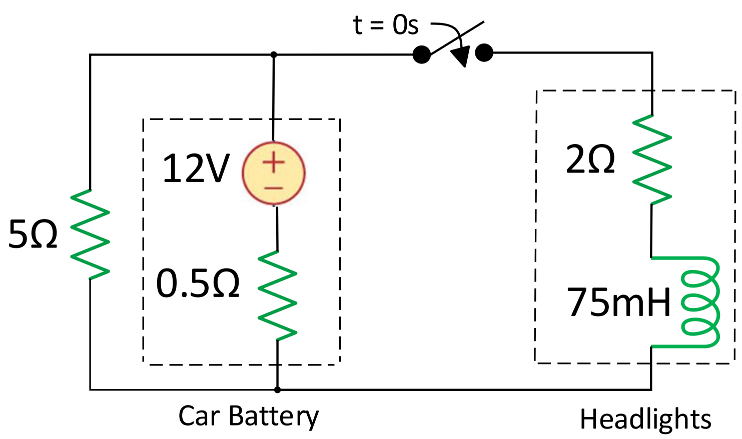

Figure 4

QUESTION 3 [25 marks]

a. **(13 marks)** Calculate i_0 in the op amp circuit shown in Figure 5.

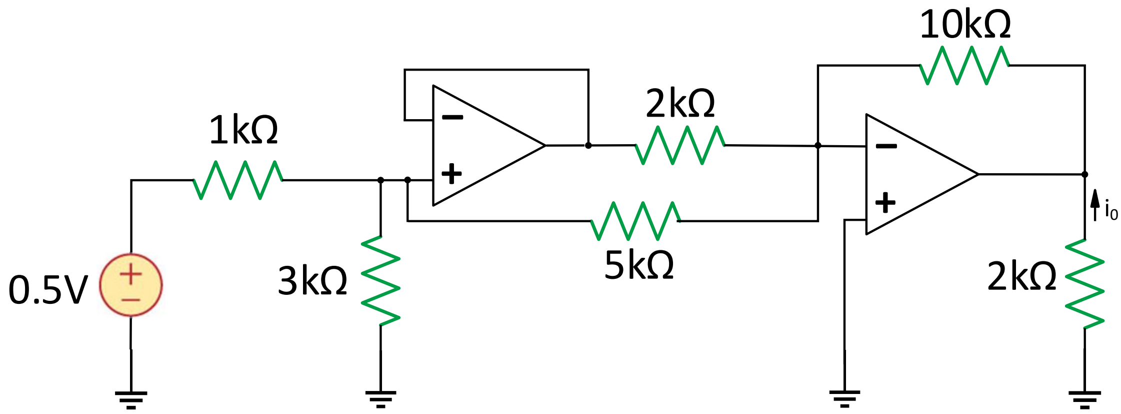

Figure 5

b. **(12 marks)** The circuit in Figure 6 represents an audio mixer, where v_1 and v_2 correspond to individual waveforms from the drums and guitar, respectively. If v_1 is a zero-phase sinusoidal signal with amplitude 240mV, v_2 is a zero-phase sinusoidal signal with amplitude 210mV, and the frequency is 60 Hz,

i. **(5 marks)** Derive an <u>expression</u> for the combined phasor voltage $\mathbf{V}_0$ in terms of the input voltages $\mathbf{V}_1$ and $\mathbf{V}_2$ (i.e., DO NOT substitute the values of $\mathbf{V}_1$ and $\mathbf{V}_2$).

ii. **(2 marks)** Substitute the values of $\mathbf{V}_1$ and $\mathbf{V}_2$ in your calculated $\mathbf{V}_0$ from part (i) and derive the combined waveform $v_0(t)$.

iii. **(5 marks)** Calculate the average power dissipated by resistor R_1 (i.e., 12kΩ resistor).

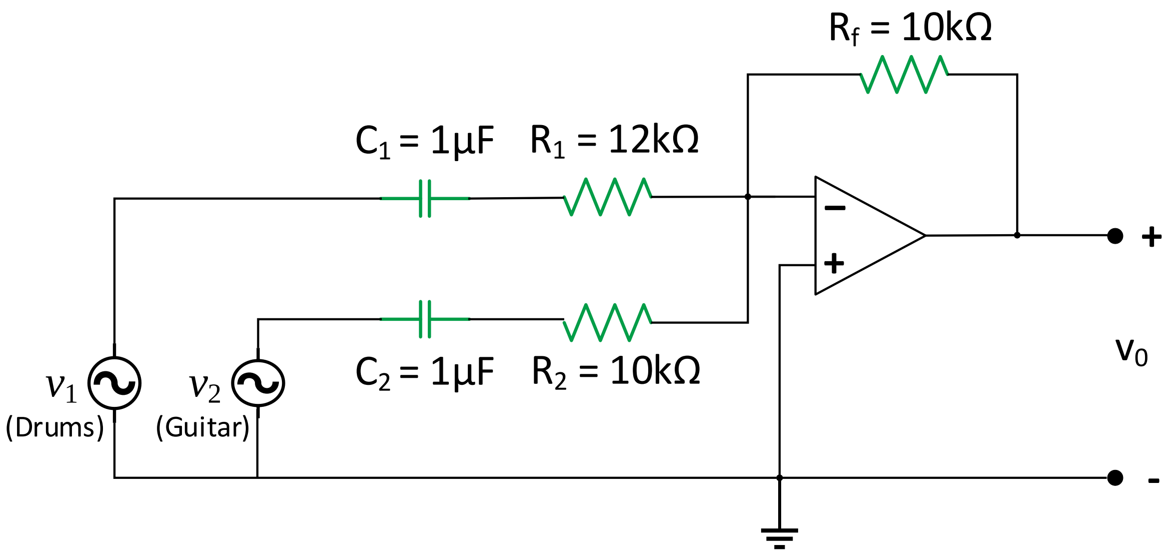

Figure 6

QUESTION 4 [23 marks]

a. **(12 marks)** The circuit shown in Figure 7 is used in a radio receiver to filter a specific interfering signal. If the amplitude of the input signal is 1V,

  i. **(5 marks)** Derive the expression of the output voltage phasor $\boldsymbol{V}_{out}$ as a function of the angular frequency ω.

  ii. **(5 marks)** Calculate the angular frequency for which the equivalent impedance of the filter between terminals a-b, $\mathbf{Z}_{eq}$, is purely resistive. Note that this angular frequency is the frequency of the interfering signal.

  iii. **(2 marks)** Substitute the value of angular frequency obtained in part (ii) in your derived output voltage phasor $\boldsymbol{V}_{out}$ from part (i) to find the value of $\boldsymbol{V}_{out}$. Based on this result, explain why the circuit shown in Figure 7 is effectively removing the interfering signal.

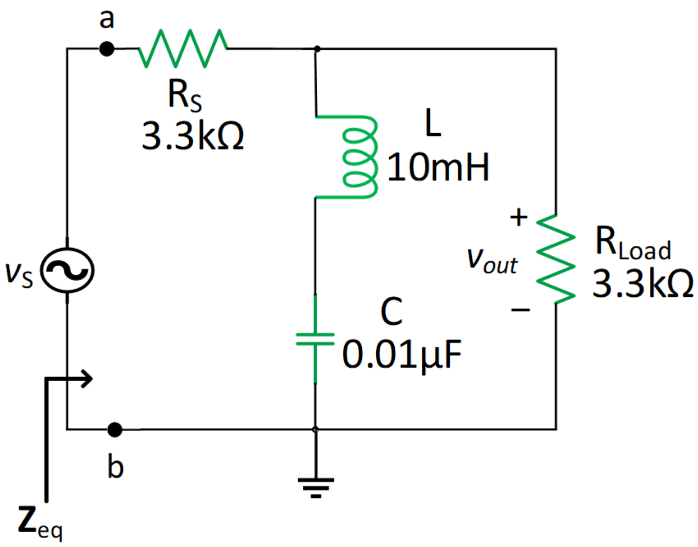

Figure 7

b. **(11 marks)** Calculate the current $i_0(t)$ in the circuit of Figure 8.

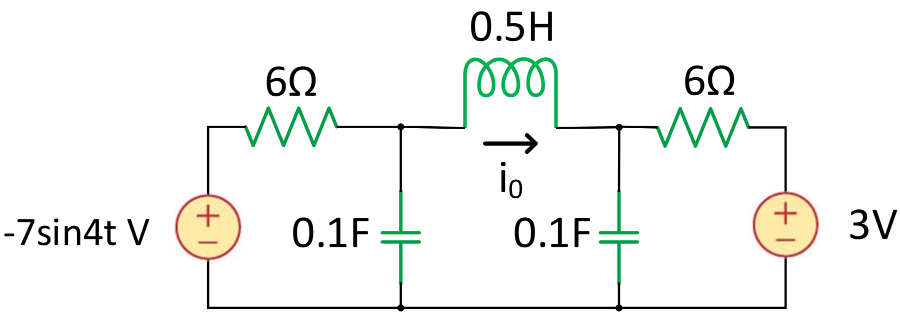

Figure 8

QUESTION 5 [12 marks]

The circuit in Figure 9 represents a household system that allows the operation of 120V, 60 Hz appliances. During winter, the owner wants to connect a heater (R_{load}) to the household system.

i. **(4 marks)** Calculate the heater load resistance R_{load} that will maximize the average power drawn from the circuit.

ii. **(8 marks)** Find the maximum average power received by the heater.

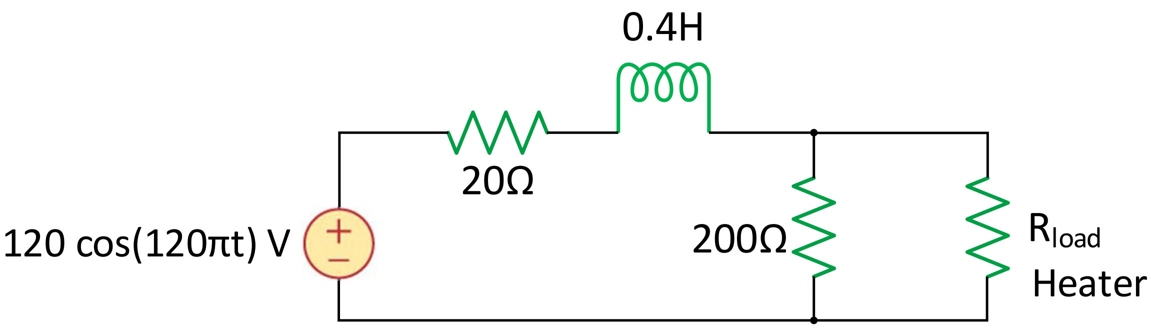

Figure 9